

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

November 2019

EC341 / EC301.01 / EC301 Electromagnetic Theory

Date: 05.11.2019, Tuesday

Time: 10.00am to 1.00pm

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.

SECTION-I

Q-1 Describe the spherical coordinate system. Carry out the conversion of Spherical coordinate system to Cartesian coordinate System. [07]

Q-2 **Attempt Following Questions** [14]

- (a) The polarization within a region having relative permittivity of 2.7 has the uniform value $\bar{P} = -0.2\bar{a}_x + 0.7\bar{a}_y + 0.3\bar{a}_z$ $\mu\text{C} / \text{m}^2$. Find (i) Electric Field Intensity **E** (ii) Electric Field Density **D** (iii) the magnitude of the voltage gradient. [07]
- (b) Find the magnitude of the current density in a conductor if (i) the electric field intensity is 0.09 V/m, the Volume charge density is $-2 \times 10^{10} \text{ C/m}^3$ and the electron mobility is $2.5 \times 10^{-3} \text{ m}^2/\text{V s}$. (ii) There are 5×10^{28} conduction electrons / m^3 and the electron drift velocity is 0.05 mm/s (c) $E = 0.05 \text{ V/m}$ and the resistivity is $2 \times 10^{-8} \Omega \text{ m}$. [07]

OR

- (a) Describe conductor properties and boundary conditions using neat sketch. [07]
- (b) Explain nature of dielectric materials. [07]

Q-3 **Write any Two** [14]

- (a) Derive Helmholtz equation in Free Space. [07]
- (b) Describe Wave Power. [07]
- (c) Describe following quantities: (a) Standing Wave Ratio (b) Skin Effect and Skin depth [07]

SECTION – II

Q-4 **Attempt Following questions** [07]

- (a) List out Maxwell's time varying equations in integral form and point form. [03]
- (b) List out Maxwell's static equations in integral and point form. [04]

Q-5 **Attempt Following questions** [14]

- (a) For the given potential $V = \frac{10}{r^2} \sin \theta \cos \phi$. [07]
- (i) Find the electric flux density **D** at $(2, \frac{\pi}{2}, 0)$.

- (ii) Calculate the work done in moving a $10 \mu C$ charge from point $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$.
- (b) Evaluate the Stokes' theorem for the the field $\vec{H} = 6xy\vec{a}_x - 3y^2\vec{a}_y \frac{A}{m}$ and the [07]
 rectangular path around the region, $2 \leq x \leq 5, -1 \leq y \leq 1$ and $z = 0$. Let the
 positive direction of $\vec{ds}$ be a_z .
- OR**
- (a) Verify the divergence theorem for the given field $\vec{A} = 2xy\vec{a}_x + x^2\vec{a}_y$ and the [07]
 rectangular parallelepiped formed by the planes $x = 0, x = 1, y = 0, y = 2, z =$
 $0, z = 3$.
- (b) State and Derive following theorems through examples. [07]
 (i) Divergence theorem
 (ii) Stokes' theorem.

Q-6 Attempt Following Questions [14]

- (a) Explain the Biot-Savart law and derive the magnetic field intensity due to [07]
 infinite long current filament.
- (b) Derive the electric field intensity ($\vec{E}$) for infinite sheet charge. [07]
- OR**
- (a) (i) Define gradient of scalar entity through example [07]
 (ii) Justify that the work done around the closed path is zero.
 (iii) Define the potential of equipotential surfaces
- (b) State the following law: [07]
 (i) Coulomb's law
 (ii) Gauss law
 (iii) Biot-Savart law